MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW POWER SCHOTTKY TTL,
BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS,
CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/328B, dated 23 March 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.3, Thermal resistance, delete and substitute:

"Cases A, C, D, R, S, L - - - - - - - - - - - - - (See MIL-M-38510, Appendix C)
Case 2 - 60°C/W 2/"

1.3, Junction temperature, delete and substitute:

"Junction temperature (T_J) 3/ - - - - - - - - - +175°C"

After footnote 1/, add:

"2/ When a thermal resistance value is included in MIL-M-38510, Appendix C, it shall supersede the value stated herein.

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 2

1.4, Normalized fanout: Delete "2/" and substitute "4/".

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

Bottom of page: Delete "2/" and substitute "4/".

PAGE 3

TABLE I, Low-level input current, I_{IL}, device types 01, 02, 03, max limits: Delete "-200" and substitute "-240".

TABLE I, Low-level input current, I_{IL}, device types 04, 05, max limits: Delete "-400" and substitute "-200".

The attached insertable replacement pages listed below are replacements for stipulated pages. When new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
13	Reprinted without change
14	14

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PAGE 5

TABLE II, MIL-STD-883 test requirements column, delete "(pre burn-in)".

3.6, second sentence: Delete in its entirety.

4.2a(1), after condition D, add "or E".

PAGE 17

TABLE III, subgroup 1, test number 50, under pin 1: Delete """ and leave open. Make this same change to test number 60.

PAGE 18

TABLE III, subgroup 1, test numbers 85 through 91, minimum limits column: Add "-40".

PAGE 24

* TABLE III, I_{IL}, maximum limits: Delete "-240" and substitute "-200 2/". This reverses a change made by amendment 3.

* TABLE III, $\overline{G}$ pin, test 99 through 115: Delete "GND" and substitute "5.5 V". For $\overline{G}$ pin, test 116, "0.4 V" is correct. This reverses a change made by amendment 3.

* TABLE III, $\overline{G}$ pin, test 117 through 133: Delete "GND" and substitute "5.5 V". For $\overline{G}$ pin, test 134, "2.7 V" is correct. This reverses a change made by amendment 3.

PAGE 25

* TABLE III, $\overline{G}$ pin, test 135 through 151: Delete "GND" and substitute "5.5 V". Test 152, $\overline{G}$ pin, "5.5 V" is correct. This reverses a change made by amendment 3.

* TABLE III, $\overline{G}$ pin, test 169: Delete "5.5 V" and substitute "GND". This reverses a change made by amendment 3.

TABLE III, $\overline{G}$ pin, test 170: Delete " " " and substitute "GND".

PAGE 27

Footnote 1/: Following "circuit", add "B and".

Footnote 2/: Delete in its entirety.

* TABLE III, footnote 2/ add: "I_{IL} limits for circuit E are -5 µA minimum, -240 µA maximum".

PAGE 29

TABLE III, subgroup 1, test numbers 72 through 92, minimum limits column, add "0"; maximum limits column, add "-200".

PAGE 30

TABLE III, subgroup 1, test numbers 137 through 152, maximum limits column: Delete "20" and substitute "-400".

TABLE III, subgroup 1, test numbers 153 through 168, maximum limits column: Delete "-400" and substitute "20".

PAGE 37

TABLE III, subgroup 1, test numbers 72 through 92, minimum limits column: Add "0". Maximum limits column: Add "-200".

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PAGE 38

TABLE III, subgroup 1, test numbers 137 through 152, maximum limits column: Delete "20" and substitute "-400".

TABLE III, subgroup 1, test numbers 153 through 168, maximum limits column: Delete "-400" and substitute "20".

PAGE 43

Footnotes: Delete existing "4/" and substitute "5/"; delete existing "5/" and substitute "4/".

PAGE 44

4.4.3c(1), after test condition D, add "or E".

4.4.3c(3): Delete "appendix B of MIL-M-38510 and".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

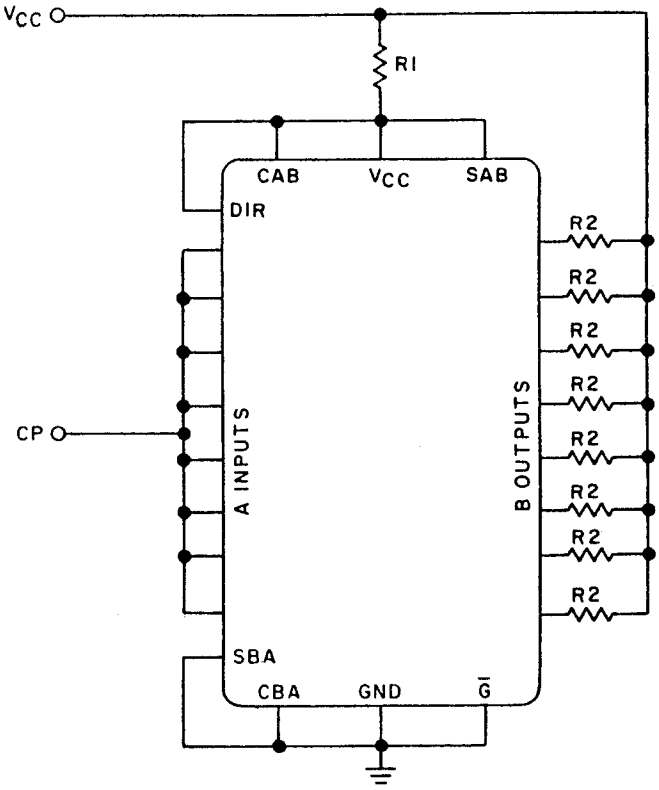
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1019)

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Device types 04 and 05

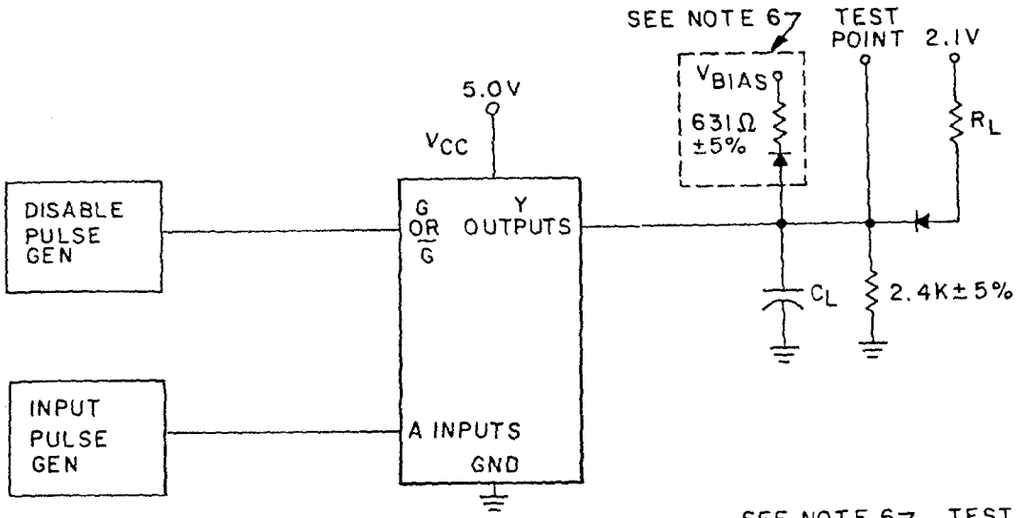


NOTES:
1. V_{CC} and R1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R2 = 470Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.5 V; V_{IH} = 2.0 V to 5.5 V.

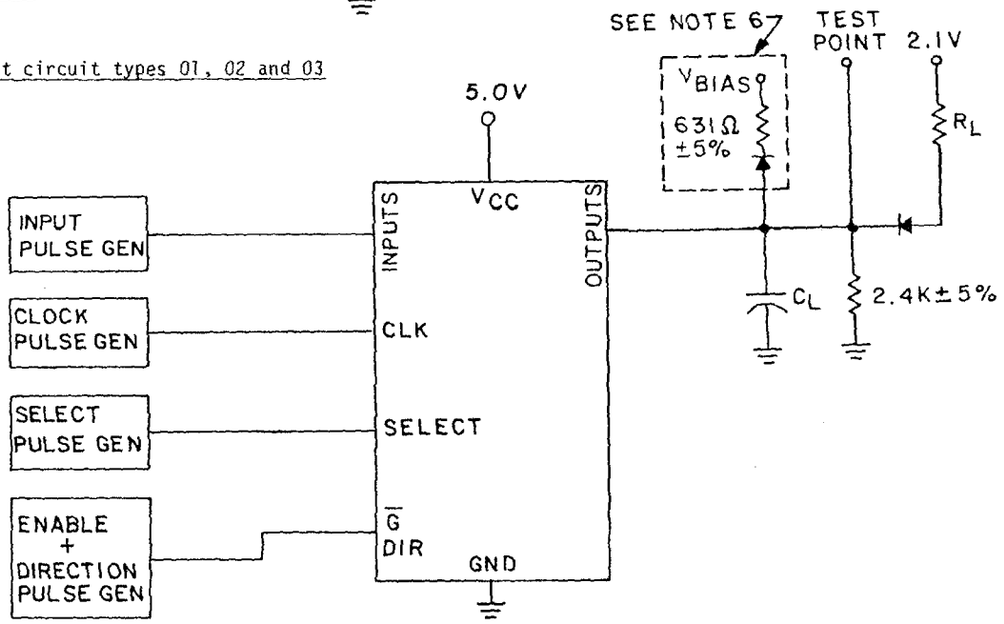
FIGURE 3. Burn-in and life test circuit - Continued.

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Test circuit types 01, 02 and 03



TEST CIRCUIT TYPES 04 and 05

NOTES:
1. R_L = 110Ω ±5%.
2. All diodes are 1N3064 or equivalent.
3. C_L = 50 pF ±10% including probe and jig capacitance.
4. The pulse generators have the following characteristics:
   V_{gen} = 3 V, PRR ≤ 1 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, Z_{OUT} = 50Ω.
5. Clock pulse characteristics: t_p(CLK) = 20 ns, t_{setup} = 20 ns.
6. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.

FIGURE 4. Switching time test circuit and waveforms.

Supersedes page 14 of amendment 3 of
4 February 1986.

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